

- No. of Printed Pages : 4
Roll No.

Roll No.

Time : 3 Hrs.

M.M. : 100

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which operation is NOT typically performed with a single point cutting tool?
- a) Turning
 - b) Shaping
 - c) Drilling
 - d) Planning
- Q.2 The speed of the lathe is controlled by:
- a) Feed mechanism
 - b) Gear Box or cone pulleys
 - c) Tailstock
 - d) Cross slide
- Q.3 In lathe operations, the depth of cut is defined as:
- a) Distance tool moves per revolution
 - b) Perpendicular distance from machined surface to uncut surface
 - c) Distance work-piece moves
 - d) Diameter of work-piece

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 What is turning? Explain principle of turning. Describe the various operations performed on a Lathe machine.
- Q.37 Describe the function of the knee, saddle and table in a column and knee milling machine.
- Q.38 Define reaming, drilling, boring, shaping, planing and slotting operations and explain their purposes.

- Q.4 The face plate is typically used for:
- Irregular shaped jobs
 - Threading
 - Boring
 - Form turning
- Q.5 In up milling, the cutter rotates:
- In the direction of feed
 - Opposite to the direction of feed
 - Randomly
 - Perpendicular to feed
- Q.6 The primary function of a milling machine is to :
- Cut, Shape and profile materials using a rotating cutter
 - Melt metal work pieces
 - Weld joints
 - Temper steel
- Q.7 Tapping is used to
- Enlarge holes
 - Create internal screw Threads
 - Drill holes
 - Smooth the hole surface
- Q.8 The twists along the body of the drill for chip removal are called
- Threads
 - Ribs
 - Helices
 - Flutes
- Q.9 During which stroke does a Shaper machine perform cutting?
- Return stroke
 - Forward stroke
 - Both A & B
 - None of these

- Q.10 What is drip oil feed lubrication?
- Oil is sprayed continuously
 - Oil drops slowly onto parts at regular intervals
 - Oil sprayed as mist
 - Grease injection

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 What is the function of a boring tool?
- Q.12 Name two types of chucks used in a lathe.
- Q.13 Describe taper turning.
- Q.14 Define depth of cut.
- Q.15 The cutting edges of a milling cutter are called flutes. (True / False)
- Q.16 Face milling is used to produce flat surfaces perpendicular to the axis of the cutter (True / False)
- Q.17 The manufacturing operation that produces high-precision holes is _____.
- Q.18 _____ machines are more suitable for machining large jobs.
- Q.19 In a slotter machine the ram reciprocates in a _____ direction.
- Q.20 Cutting fluids help improve tool _____ and surface finish.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What are the Advantages and disadvantages of tipped single point cutting tools compared to solid tools?
- Q.22 How do positive and negative rake angles influence cutting forces and tool life?